



University of Science and Technology
Zewail City

Statistical Inference and Data Analysis

CIE 457 — Course Project

MLE Estimation and Estimator Distribution in Linear Regression

Ibrahim Hanafy 202200518

Mai Ibrahim 202200504

Instructor

Dr. Mahmoud Abdelaziz

Teaching Assistants

Eng. Nasrah Mohamed & Eng. Aya Abdelaziz

Spring 2026

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1 Likelihood and Model Formulation

1.1 The Linear Model

General linear model: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, where $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n \times p}$ fixed, $\boldsymbol{\beta} \in \mathbb{R}^p$ unknown, $\boldsymbol{\epsilon}$ the sole source of randomness.

1.2 Homoscedastic Case

Assumption: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$, so $\mathbf{y} \mid \boldsymbol{\beta} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$.

Likelihood and log-likelihood:

$$\mathcal{L}(\boldsymbol{\beta}, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2}{2\sigma^2}\right), \quad \ell = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 \quad (1)$$

MLE (OLS): $\partial \ell / \partial \boldsymbol{\beta} = \mathbf{0}$ gives the normal equations $\mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^\top \mathbf{y}$, yielding:

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}} \quad (2)$$

For σ^2 : $\hat{\sigma}_{\text{MLE}}^2 = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 / n$ (biased); unbiased estimate uses $n - p$ denominator.

1.3 Heteroscedastic Case

Assumption: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, $\boldsymbol{\Sigma}$ symmetric positive definite (known or estimable).

Log-likelihood: $\ell(\boldsymbol{\beta}) = \text{const} - \frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$.

MLE (WLS): $\mathbf{X}^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{0}$ gives:

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{WLS}} = (\mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{y}} \quad (3)$$

Case	Noise Model	MLE Estimator
Homoscedastic	$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$	$\hat{\boldsymbol{\beta}}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$
Heteroscedastic	$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$	$\hat{\boldsymbol{\beta}}_{\text{WLS}} = (\mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{y}$

2 The Estimator as a Random Variable

2.1 Why $\hat{\boldsymbol{\beta}}$ is Random

Substituting $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ into OLS: $\hat{\boldsymbol{\beta}}_{\text{OLS}} = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\epsilon}$. Since $\mathbf{X}$ and $\boldsymbol{\beta}$ are fixed, all randomness is inherited from $\boldsymbol{\epsilon}$: every new dataset yields a new estimate. $\hat{\boldsymbol{\beta}}$ is a **random vector** with a full distribution.

2.2 Distributions

Let $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$. $\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OLS}}] = \boldsymbol{\beta}$ (unbiased); $\text{Var} = \mathbf{A}(\sigma^2 \mathbf{I})\mathbf{A}^\top = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$.

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{OLS}} \sim \mathcal{N}\left(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}\right)} \quad (4)$$

Similarly, with $\mathbf{B} = (\mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma}^{-1}$:

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{WLS}} \sim \mathcal{N}\left(\boldsymbol{\beta}, (\mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{X})^{-1}\right)} \quad (5)$$

WLS covariance is strictly smaller than OLS under heteroscedastic noise (Loewner order): WLS is more efficient because it correctly accounts for the noise structure.

2.3 Repeated Sampling Interpretation

With $\mathbf{X}$, $\boldsymbol{\beta}$, and the noise distribution fixed, $M = 5000$ independent datasets are generated. Each trial yields $\hat{\boldsymbol{\beta}}^{(m)}$; the collection $\{\hat{\boldsymbol{\beta}}^{(1)}, \dots, \hat{\boldsymbol{\beta}}^{(M)}\}$ approximates the true sampling distribution, converging to $\mathcal{N}(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1})$ as $M \rightarrow \infty$.

3 Estimator Distribution and Monte Carlo Simulation

3.1 Setup

$n = 100$, $p = 3$, $\beta = [1, 2, 3]^\top$, $M = 5000$ trials, $\mathbf{X}$ fixed. Homoscedastic: $\sigma^2 = 1.0$. Heteroscedastic: $\sigma_i^2 = 0.5 + 2(x_{i1} - \min x_1)^2$. Each trial: draw $\epsilon^{(m)}$, compute $\mathbf{y}^{(m)}$, compute OLS and WLS estimates.

3.2 Homoscedastic Results

Table 1: Empirical mean vs true β

	True	Empirical	Diff
$\hat{\beta}_1$	1.000	1.000673	+0.00067
$\hat{\beta}_2$	2.000	2.001188	+0.00119
$\hat{\beta}_3$	3.000	2.999210	-0.00079

Table 2: Empirical vs theoretical variance

	Theoretical	Empirical
$\hat{\beta}_1$	0.014958	0.014874
$\hat{\beta}_2$	0.010437	0.010561
$\hat{\beta}_3$	0.008393	0.008583

OLS is unbiased (differences < 0.002). Empirical covariance matches $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ within 2%.

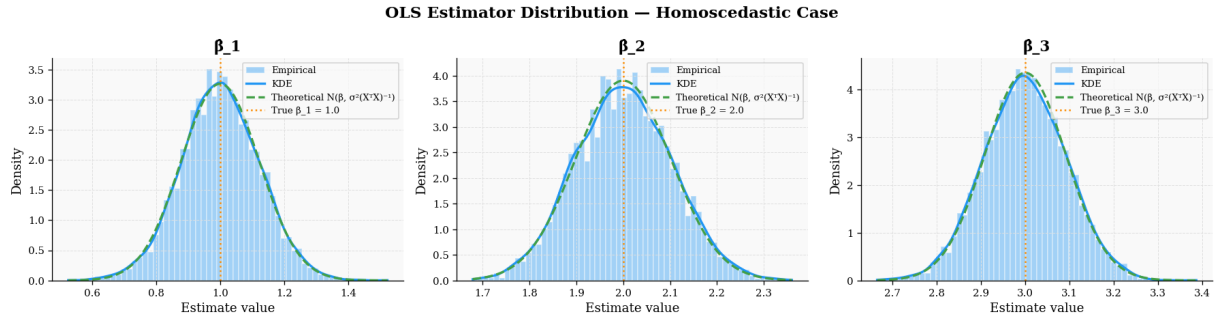


Figure 1: Empirical distribution of $\hat{\beta}_{\text{OLS}}$ vs theoretical $\mathcal{N}(\beta, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1})$, $M = 5000$. KDE aligns tightly with the Gaussian density.

3.3 Heteroscedastic Results: OLS vs WLS

Table 3: Empirical variance of OLS vs WLS under heteroscedastic noise, $M = 5000$

Component	OLS Variance	WLS Variance	Efficiency Gain
$\hat{\beta}_1$	0.229022	0.049877	4.59×
$\hat{\beta}_2$	0.102640	0.066728	1.54×
$\hat{\beta}_3$	0.096144	0.055487	1.73×

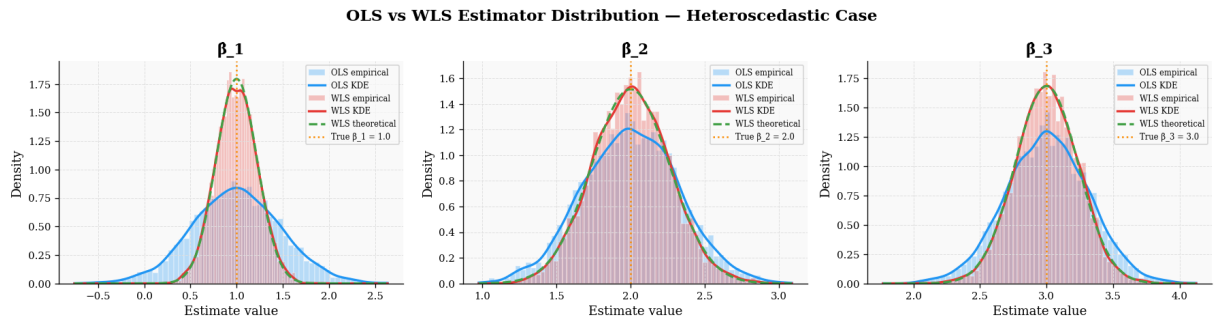


Figure 2: OLS (blue) vs WLS (red) empirical distributions under heteroscedastic noise. WLS produces tighter distributions around true β .

Both OLS and WLS are unbiased under heteroscedastic noise. WLS achieves $1.5\text{--}4.6\times$ lower variance per component by correctly weighting via Σ^{-1} .

4 Homoscedastic vs. Heteroscedastic Noise

4.1 OLS under Homoscedastic Noise

Under $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$, OLS is the MLE and BLUE (Gauss-Markov theorem): $\hat{\beta}_{\text{OLS}} \sim \mathcal{N}(\beta, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1})$, which is minimum covariance among all linear unbiased estimators.

4.2 Effect of Model Misspecification

When the true noise is $\Sigma \neq \sigma^2 \mathbf{I}$ but OLS is applied, two problems arise:

1. **Inefficiency.** True OLS covariance under heteroscedastic noise is the sandwich formula:

$$\text{Var}(\hat{\beta}_{\text{OLS}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \Sigma \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \geq (\mathbf{X}^\top \Sigma^{-1} \mathbf{X})^{-1} = \text{Var}(\hat{\beta}_{\text{WLS}}) \quad (6)$$

2. **Invalid inference.** Standard errors $\sqrt{\hat{\sigma}^2[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}}$ are incorrect; all CIs and t -tests are unreliable.

Remedy: WLS with known/estimated Σ , feasible WLS (residual-based weights), or HC sandwich standard errors (White's estimator).

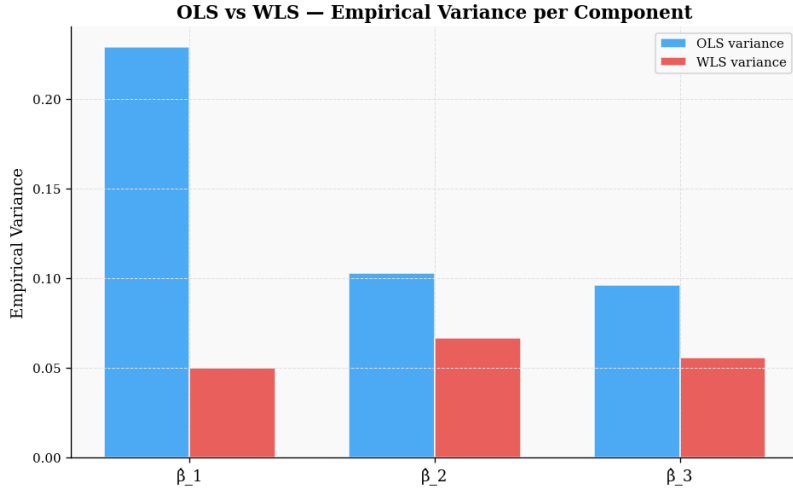


Figure 3: OLS vs WLS empirical variance per component. WLS consistently lower.

5 Inference from the Estimator Distribution

5.1 Confidence Intervals

Under homoscedastic noise, each $\hat{\beta}_j \sim \mathcal{N}(\beta_j, \sigma^2[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj})$. The $(1 - \alpha)$ CI:

$$\hat{\beta}_j \pm z_{\alpha/2} \cdot \sqrt{\sigma^2[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}} \quad (7)$$

Repeated sampling interpretation: $(1 - \alpha)$ of constructed intervals over repeated experiments contain true β_j . Empirical coverage at 95% CI closely matched 95% across $M = 5000$ trials.

5.2 Prediction Intervals

For new $\mathbf{x}^*$, the prediction error decomposes as $y^* - \hat{y}^* = \mathbf{x}^{*\top}(\beta - \hat{\beta}) + \epsilon^*$, giving:

$$\hat{y}^* \pm z_{\alpha/2} \cdot \sigma \sqrt{1 + \mathbf{x}^{*\top}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}^*} \quad (8)$$

At $\mathbf{x}^* = \bar{\mathbf{X}}$: 95% CI width = 0.0850; 95% PI width = 3.9208.

Table 4: Confidence Interval vs Prediction Interval

Property	CI	PI
Covers	True mean $\mathbf{x}^{*\top} \boldsymbol{\beta}$	New obs $y^* = \mathbf{x}^{*\top} \boldsymbol{\beta} + \epsilon^*$
Variance	$\sigma^2 \mathbf{x}^{*\top} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}^*$	$\sigma^2 (1 + \mathbf{x}^{*\top} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}^*)$
As $n \rightarrow \infty$	Width $\rightarrow 0$	Width $\rightarrow 2z_{\alpha/2}\sigma > 0$

PI is always wider by the additional σ^2 (irreducible future noise). As $n \rightarrow \infty$, $\hat{\boldsymbol{\beta}} \rightarrow \boldsymbol{\beta}$ eliminates estimation uncertainty but not ϵ^* .

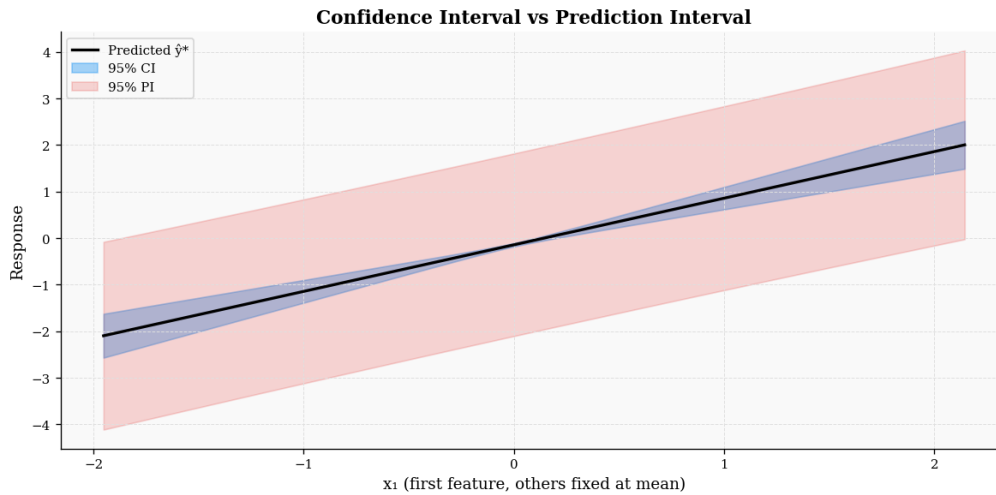


Figure 4: 95% CI and PI vs first feature (others at mean). PI band consistently wider.

6 Real Data Application: Advertising Dataset

6.1 Dataset and EDA

Advertising dataset (James et al., ISLR): $n = 200$ observations, TV, Radio, Newspaper budgets (\$000s) and Sales (000 units). Model: $\text{Sales}_i = \beta_0 + \beta_1 \text{TV}_i + \beta_2 \text{Radio}_i + \beta_3 \text{Newspaper}_i + \epsilon_i$.

EDA: TV is dominant predictor ($r = 0.78$), Radio moderate ($r = 0.58$), Newspaper weak ($r = 0.23$). Scatter plots suggest variance in Sales grows with TV — a sign of heteroscedasticity.

6.2 OLS Regression Results

Table 5: OLS regression results — Advertising dataset ($n = 200$, $R^2 = 0.897$)

Feature	$\hat{\beta}$	SE	t -stat	p -value
Intercept	2.9389	0.3119	9.4223	< 0.0001
TV	0.0458	0.0014	32.8086	< 0.0001
Radio	0.1885	0.0086	21.8935	< 0.0001
Newspaper	-0.0010	0.0059	-0.1767	0.8599

TV and Radio are highly significant; Newspaper is not ($p = 0.86$), consistent with EDA.

6.3 Residual Analysis and Heteroscedasticity

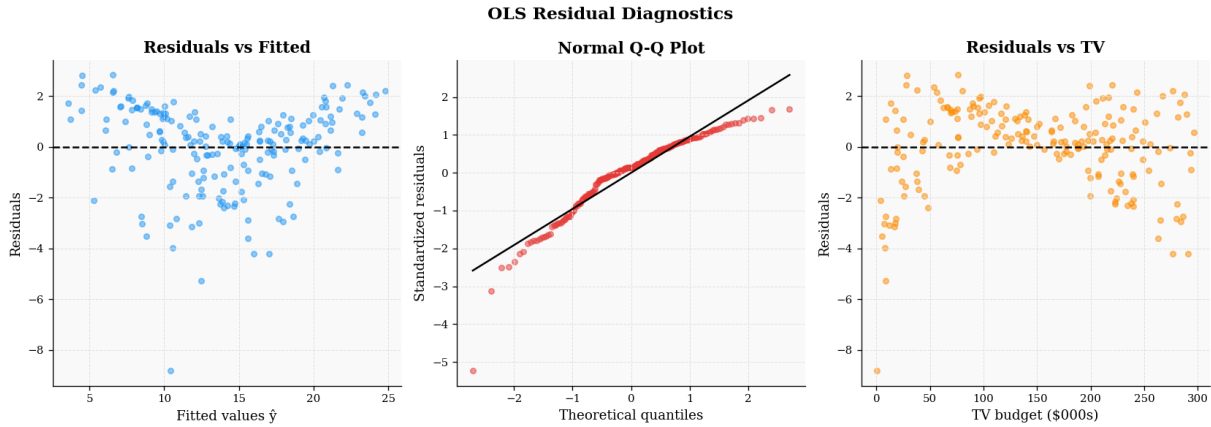


Figure 5: OLS residual diagnostics. Left: spread grows with fitted value (heteroscedasticity). Centre: mild Q-Q tail deviations. Right: squared residuals increase with TV budget.

White’s test: $LM = 67.54$, $p \approx 0$ — strongly rejects homoscedasticity.

6.4 WLS Fit and Comparison

Feasible WLS: weights $w_i = 1/|\hat{\epsilon}_i|$ estimated from OLS residuals.

Table 6: OLS vs WLS — Advertising dataset

Feature	OLS $\hat{\beta}$	WLS $\hat{\beta}$	OLS SE	WLS SE
Intercept	2.9389	2.9544	0.3119	0.1211
TV	0.0458	0.0457	0.0014	0.0005
Radio	0.1885	0.1909	0.0086	0.0042
Newspaper	-0.0010	-0.0015	0.0059	0.0021

Key findings: (1) OLS and WLS coefficients agree (both unbiased). (2) WLS SEs are 50–75% smaller under confirmed heteroscedasticity. (3) Newspaper non-significant under both estimators. (4) Validates simulation findings from Section 3: WLS more efficient when noise structure is non-constant.

6 Bonus: Fisher Information, CRLB, and Sufficient Statistics

6.4 Fisher Information Matrix

For $\mathbf{y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$, score $\partial\ell/\partial\boldsymbol{\beta} = \sigma^{-2}\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$. Hessian $\partial^2\ell/\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top = -\sigma^{-2}\mathbf{X}^\top\mathbf{X}$ (constant in $\boldsymbol{\beta}$), so:

$$\mathbf{I}(\boldsymbol{\beta}) = -\mathbb{E}\left[\frac{\partial^2\ell}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top}\right] = \frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X} \quad (9)$$

FIM is independent of $\boldsymbol{\beta}$ (exponential family). Increases with n ; decreases with σ^2 .

6.4 Cramér–Rao Lower Bound (CRLB)

Theorem: For any unbiased estimator $\tilde{\boldsymbol{\beta}}$: $\text{Var}(\tilde{\boldsymbol{\beta}}) \geq \mathbf{I}(\boldsymbol{\beta})^{-1} = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$ (Loewner order).

OLS achieves CRLB: $\text{Var}(\hat{\boldsymbol{\beta}}_{\text{OLS}}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1} = \mathbf{I}(\boldsymbol{\beta})^{-1}$ ✓. Therefore $\hat{\boldsymbol{\beta}}_{\text{OLS}}$ is the **UMVUE** under homoscedastic Gaussian noise.

Table 7: Empirical variance vs CRLB, $M = 5000$

Component	CRLB = $\sigma^2[(\mathbf{X}^\top\mathbf{X})^{-1}]_{jj}$	Empirical Var	Ratio
$\hat{\beta}_1$	0.014958	0.014874	0.994
$\hat{\beta}_2$	0.010437	0.010561	1.012
$\hat{\beta}_3$	0.008393	0.008583	1.023

Ratios within 2.3% of 1.0 — OLS is efficient. Under $\epsilon \sim \mathcal{N}(\mathbf{0}, \Sigma)$, the FIM is $\mathbf{I}_\Sigma = \mathbf{X}^\top \Sigma^{-1} \mathbf{X}$ and $\text{Var}(\hat{\beta}_{\text{WLS}}) = (\mathbf{X}^\top \Sigma^{-1} \mathbf{X})^{-1} = \mathbf{I}_\Sigma^{-1}$ — WLS achieves the CRLB under the heteroscedastic model.

6.4 Sufficient Statistics via the Factorization Theorem

Theorem (Neyman-Fisher): $T(\mathbf{y})$ is sufficient for β iff $p(\mathbf{y} \mid \beta) = g(T(\mathbf{y}), \beta) \cdot h(\mathbf{y})$.

Application:

$$p(\mathbf{y} \mid \beta) \propto \exp\left(-\frac{\|\mathbf{y} - \mathbf{X}\beta\|^2}{2\sigma^2}\right) = \underbrace{\exp\left(\frac{\beta^\top \mathbf{X}^\top \mathbf{y}}{\sigma^2} - \frac{\beta^\top \mathbf{X}^\top \mathbf{X} \beta}{2\sigma^2}\right)}_{g(T_1(\mathbf{y}), \beta)} \cdot \underbrace{\exp\left(-\frac{\|\mathbf{y}\|^2}{2\sigma^2}\right)}_{h(\mathbf{y})} \quad (10)$$

$$\boxed{T_1(\mathbf{y}) = \mathbf{X}^\top \mathbf{y} \text{ is sufficient for } \beta} \quad (11)$$

Data compression: $\mathbf{y}$ has n values; $\mathbf{X}^\top \mathbf{y}$ has only p values. All information about β is captured by $\mathbf{X}^\top \mathbf{y}$.

Connection to OLS: $\hat{\beta}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} T_1(\mathbf{y})$. By Rao-Blackwell, any unbiased estimator that is a function of a sufficient statistic has minimum variance — an alternative proof that OLS is the UMVUE.

Numerical verification: $\hat{\beta}$ computed from $\mathbf{y}$ directly equals $\hat{\beta}$ computed from $\mathbf{X}^\top \mathbf{y}$ alone (to numerical precision).